A Multi-Hazard Assessment of 2,696 US Data Centers

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Background

- AI and cloud demand has concentrated US computing into a few dozen metro clusters that face earthquakes, wildfire and flooding.
- Existing sources are proprietary, or give city-level coordinates never checked against a building.
- We built an open, building-level inventory of the contiguous US, with positional uncertainty carried through the hazard sampling.
- We measured exposure, meaning whether a site lies in a hazard zone. We did not model damage, so nothing here is a loss estimate.

Limitations

- Coverage is uneven. OpenStreetMap supplies 83% of Virginia's records but 48% of California's, so state contrasts partly reflect who mapped them.
- Hail and wind reports rise with population density (Spearman 0.39 and 0.34 within state), so we excluded both.
- No open source lists power capacity, so a small suite counts the same as a large campus.
- 40 sites shown as clear sit outside FEMA's mapped extent and 8 outside the wildfire raster, so 35% is a floor.

Materials and Methods

We merged OpenStreetMap, PeeringDB and Wikidata into one row per site across the contiguous US, logging all 194 merges with their evidence.

Every coordinate was checked against FEMA USA Structures footprints. 1,681 fall inside a building, 993 match the nearest, 22 have no match. Wildfire buffers use an equal-area projection, the rest are point queries:

Earthquake	USGS ASCE 7-22, MCE_G PGA, site class BC
Wildfire	USFS Hazard Potential 270 m, max in 2.4 km
Flood	FEMA NFHL layer 28, SFHA flag
Water stress	WRI Aqueduct 4.0 (a supply limit, reported apart)
Uncertainty	500 relocations per site, 10-500 m by precision

A site counts as exposed if it lies within 2.4 km of land rated High or Very High for wildfire, inside a FEMA Special Flood Hazard Area, or at MCE_G peak ground acceleration of 0.30 g or above. Where a layer has no value there, the site is recorded as unknown, not as safe.

Results

2,696 facilities mapped

35% face a mapped hazard

0.7% of Virginia's 409

100% of California's 223

Exposure is set by geography. Virginia's 409 sites are 0.7% exposed. California's 223 are 100%.

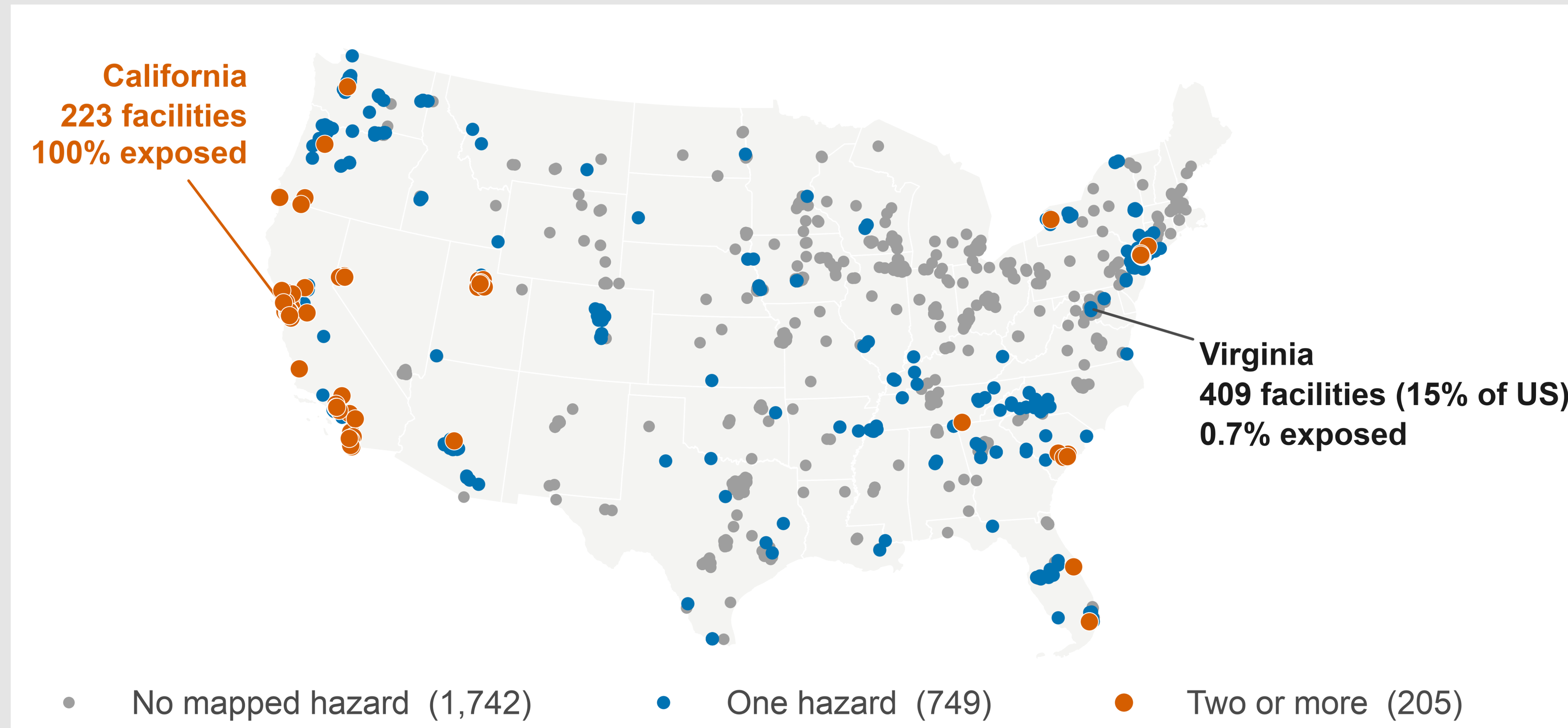

Fig 1. One third of US data centers face a mapped hazard. Sites facing two or more (orange) concentrate in California and the Northeast.

Which hazard dominates depends on the region. New Jersey reaches 100% through wildfire alone (the Pine Barrens rate High or Very High) and 0% through earthquake. California is 96% earthquake.

Nationally 646 sites are exposed to wildfire, 448 to earthquake and 71 to flood. 205 face more than one, so these overlap.

Virginia avoids the mapped hazards but not water. 31% of its 409 sites lie in high water-stress basins, against 0.7% for the three hazards.

Widening the wildfire buffer from 2.4 km to 5 km takes that count from 646 to 953, so the threshold moves the answer by half.

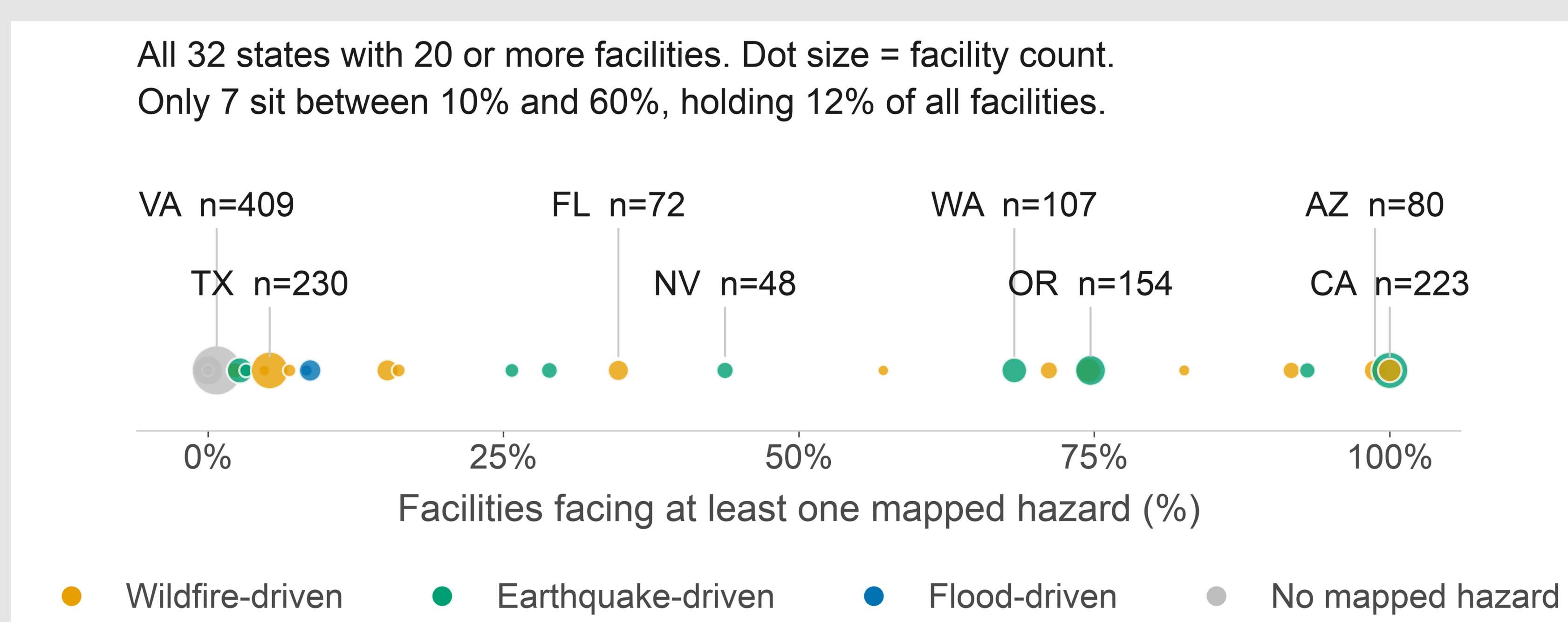

Fig 2. Every state with 20 or more facilities. 25 of 32 sit below 10% or above 60%, and colour shows which hazard drives each one.

Conclusions

- A national average hides the pattern. Only 7 of the 32 states with 20 or more sites fall between 10% and 60% exposed.
- Most sites sit in low-exposure regions, but we cannot say whether hazard influenced siting.
- New Jersey and California are both 100% exposed for opposite reasons, so a national index must report hazards separately.
- Next: fragility curves to turn exposure into estimated loss.

Two checks that came back negative

Sampling hazard across a building's whole footprint instead of its centre changed the answer for 29 of the 326 sites where both could be compared. Building height showed no association with exposure once state was held constant.

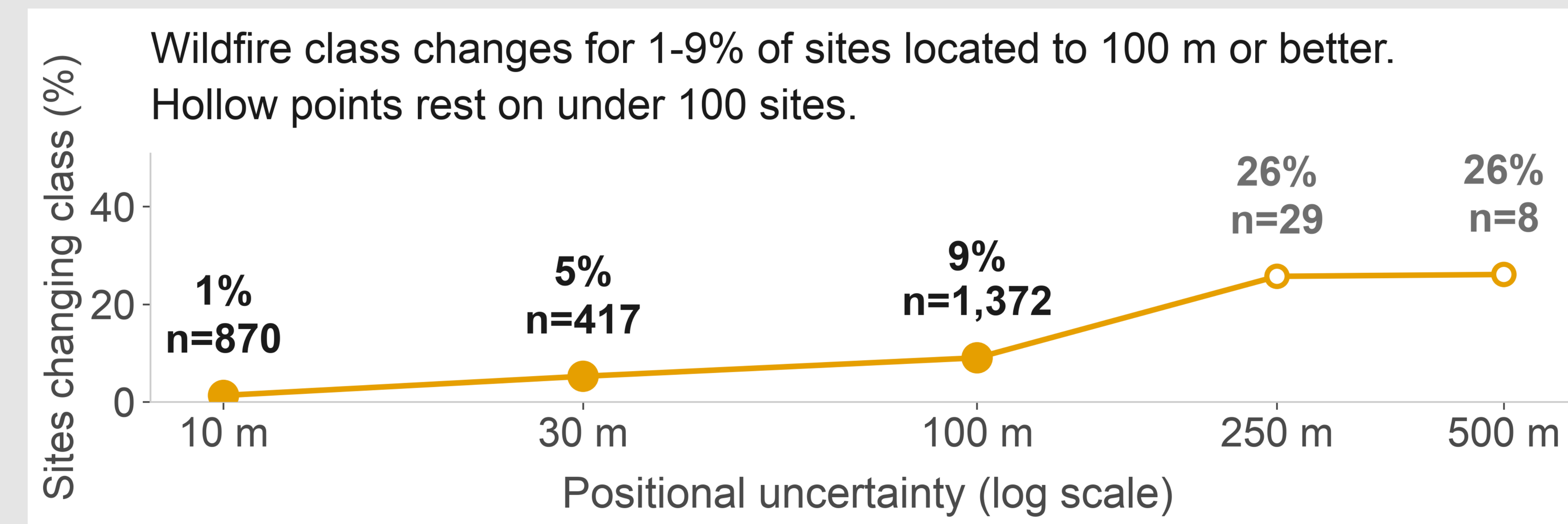

Fig 3. We re-ran the analysis with simulated location error.

Major Citations

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FEMA and ORNL (2024) USA Structures. gis.fema.gov

WRI (2023) Aqueduct 4.0 Water Risk Atlas.

Acknowledgements

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